

Heterocycles

Tunable and Cooperative Catalysis for Enantioselective Pictet-Spengler Reaction with Varied Nitrogen-Containing Heterocyclic Carboxaldehydes

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Abstract: Herein we report an organocatalytic enantioselective functionalization of heterocyclic carboxaldehydes via the Pictet-Spengler reaction. Through careful pairing of novel squaramide and Brønsted acid catalysts, our method tolerates a breadth of heterocycles, enabling preparation of a series of heterocycle conjugated β -(tetrahydro)carboline in good yield and enantioselectivity. Careful selection of carboxylic acid co-catalyst is essential for toleration of a variety of regioisomeric heterocycles. Utility is demonstrated via the three-step stereo-selective preparation of pyridine-containing analogues of potent selective estrogen receptor downregulator and U.S. FDA approved drug Tadalafil.

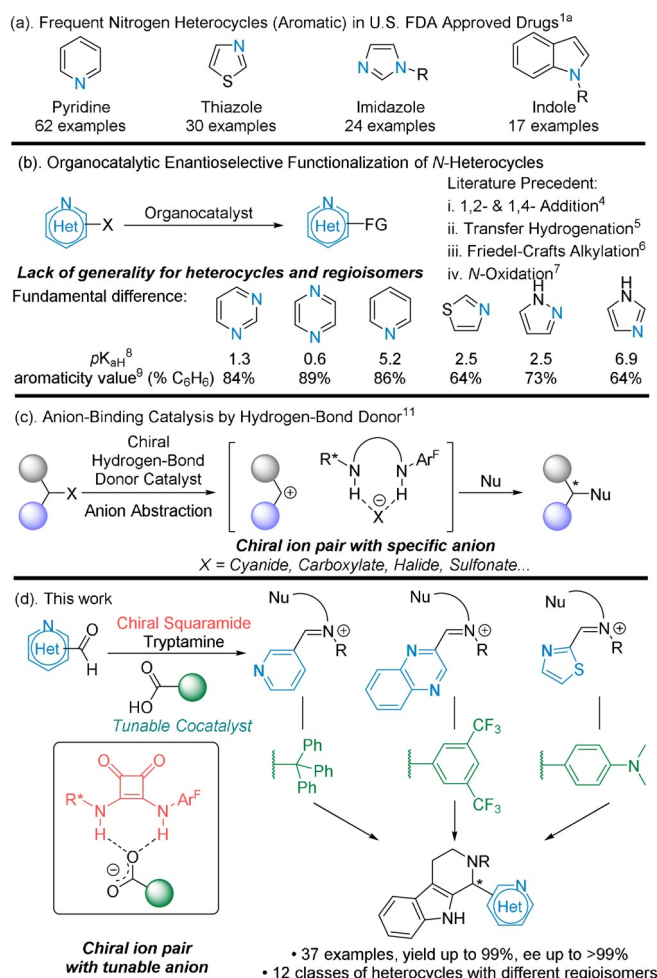
Introduction

Heterocycles play an important role in pharmaceutical development. A survey of United States Food and Drug Administration (U.S. FDA)-approved unique small molecule drugs, revealed that 59% contain at least one nitrogen heterocycle (*N*-heterocycle) in 2014 (Scheme 1a).^[1a] This proportion climbs to ca. 78% among novel drug approvals in 2020.^[1b] The prevalence of heterocyclic cores in both biologically and pharmaceutically relevant scaffolds necessitates highly effective syntheses. Effective strategies involve the rapid, stereocontrolled preparation of diverse scaffolds.^[2] Organocatalytic approaches have been utilized for facile preparation of diverse building blocks for synthesis of bioactive molecules in an often environmentally benign manner.^[3] Despite many profound advances, studies focusing on the enantioselective functionalization of nitrogen-containing aromatic heterocycles remain sparse (Scheme 1b). Venerable reactions, such as 1,2- or 1,4- additions,^[4] transfer hydrogenations,^[5] Friedel-Crafts alkylations^[6] and *N*-oxidations^[7] have all been studied in detail. Yet, many of these triumphant reports reveal lower efficiency (enantioselectivity, yield, rate), when the catalytic methodologies confront

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substrates bearing diverse *N*-heterocycles as critical moieties, limiting industrial application. This limitation might be attributed to fundamental differences among heterocycles, such as basicity (reflected by the pK_a)^[8] and electronic properties (reflected by aromaticity value);^[9] furthermore, regioisomers of heterocyclic substrates alter the relative proximity between functional group and coordinating nitrogen, resulting in additional complexity.

Anion-binding catalysis (Scheme 1c) (or asymmetric counter-anion directed catalysis) is an important field in asymmetric organocatalysis,^[10] pioneered by Jacobsen^[11] and List.^[12] Numerous advances have also been contributed by



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Toste,^[13] Seidel,^[14] Ooi,^[15] and many others.^[16] Anion-binding catalysis with hydrogen-bond donors (HBD) as catalysts have been successfully utilized with cyanide, carboxylate, halide and sulfonate for a broad range of catalytic asymmetric reactions.^[11] However, to the best of our knowledge, tuning the anion (e.g. carboxylate) to specific substrates with varied steric and electronic properties, within a reaction type, represented a new opportunity for this strategy within the context of demanding, pharmaceutically relevant functionalities.

Given the privileged bioactivity of indole alkaloids,^[17] such as β -(tetrahydro)carboline and ibogaine, efficient methodology for the stereoselective assembly of densely functionalized, stereochemically complex indole derivatives, bearing *N*-heterocycles is of high value in medicinal and process chemistry communities; this type of molecular complexity is often positively correlated with biological activity.^[18]

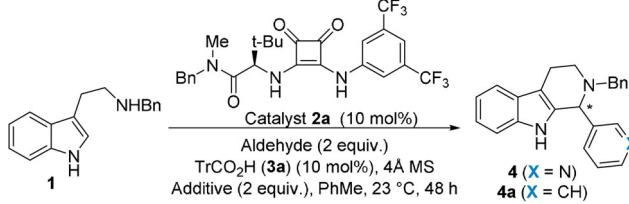
Herein we report the enantioselective functionalization of numerous and varied nitrogen heterocycles^[19] via the venerable Pictet–Spengler (PS) reaction^[20] catalyzed by a dipeptide-functionalized squaramide and carboxylic acid co-catalysts (Scheme 1d).^[21] A broad scope of heterocycles was examined, revealing critical differences in both reactivity and selectivity, not only among different heterocycles but also among regioisomers of the same *N*-heterocycle. Building on the seminal reports of Jacobsen et al.^[20f] on thiourea–Brønsted acid-co-catalyzed PS reactions, we demonstrated that matching of a squaramide hydrogen bond donor catalyst with a finely tuned achiral carboxylic acid co-catalyst is critical to achieve excellent enantioselectivity across the broadest range of heterocyclic classes. Our findings culminate in the demonstration of these catalyst systems to the rapid construction of pyridine-containing analogues of potent selective estrogen receptor downregulator (SERD)^[22] and U.S. FDA approved drug Tadalafil.^[23]

Results and Discussion

Our study began with the initial comparison between benzaldehyde and nicotinaldehyde (pyridine-3-carboxaldehyde) as aldehyde substrate in the Pictet–Spengler (PS) reaction by means of both reactivity and enantioselectivity. We observed immediately that inclusion of either trifluoroethanol or ethanol could drastically promote reactivity without significant erosion of enantioselectivity in the case of nicotinaldehyde. The role of alcohol may be hypothesized to improve catalyst turnover (Table 1).^[24] After a brief examination of reaction condition (See the Supporting Information, Table S1), we evaluated a series of hydrogen-bond donor catalysts. Initial comparison of squaramide **2a** and thiourea **2aa** revealed that both catalysts offer similar reactivity, yet a significantly higher enantioselectivity was offered by **2a** (Table 2, **2a**: 94% yield, 15:85 er; **2aa**: 92% yield, 27:73 er).

Based on these findings, a series of *tert*-leucine (Tle) derived chiral squaramides with different *C*-termini was evaluated. Increasing the steric bulk of the nitrogen substituents at the *C*-termini from methyl benzyl (**2a**) to benzhydryl

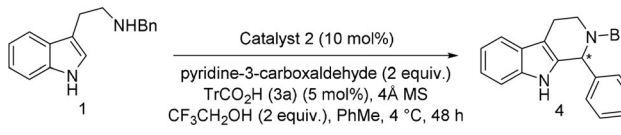
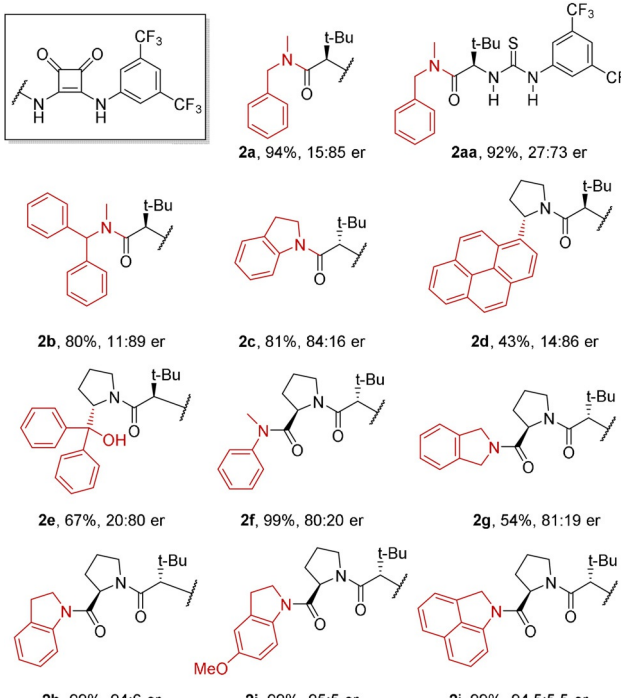
Table 1: Comparison between nicotinaldehyde and benzaldehyde.^[a]



Entry	Aldehyde	Additive	Product	Yield [%]	er
1	nicotinaldehyde	CF ₃ CH ₂ OH	4	90	18:82
2	nicotinaldehyde	none	4	55	17:83
3	benzaldehyde	CF ₃ CH ₂ OH	4a	99	75:25
4	benzaldehyde	none	4a	99	85:15
5	nicotinaldehyde	EtOH	4	87	17.5:82.5
6	nicotinaldehyde	(CF ₃) ₂ CHOH	4	85	25:75

[a] Reactions were performed at 0.05 mmol scale. Yield was determined by ¹H NMR spectroscopy with CH₂Br₂ as internal standard. Enantioselectivity (er) was measured by HPLC with chiral stationary phase and presented as order of elution from HPLC. Both **4** and **4a** have the same absolute configuration.^[20d]

Table 2: Evaluation of hydrogen-bond donor catalyst.^[a]

2a , 94%, 15:85 er	2aa , 92%, 27:73 er	
2b , 80%, 11:89 er	2c , 81%, 84:16 er	2d , 43%, 14:86 er
2e , 67%, 20:80 er	2f , 99%, 80:20 er	2g , 54%, 81:19 er
2h , 99%, 94:6 er	2i , 99%, 95:5 er	2j , 99%, 94.5:5.5 er

[a] Reactions were performed at 0.05 mmol scale. Yield was determined by ¹H NMR spectroscopy with CH₂Br₂ as internal standard. Enantioselectivity was measured by HPLC with chiral stationary phase.

(**2b**) provided a small boost to the enantioselectivity from 15:85 er to 11:89 er. However, pyrenyl (**2d**) or diphenylmethanol substituted pyrrolidine-derived catalysts (**2e**) provided little change. We envisioned that anilides at the *C*-terminus of

these Tle-Pro dipeptides might alter the nature of any π - π interactions in the transition state.^[25] Returning to C-terminal amides in the dipeptide scaffold, along with their attendant H-bonding capability, led to the dipeptide-squaramides **2f**, **2g**, **2h**, **2i** and **2j**.^[26]

While **2f** and **2g** led to moderate selectivity (80:20 er; 81:19 er, respectively), catalysts **2h**, **2i** and **2j** offered striking enantioselectivity and yield (94:6–95:5 er; near quantitative yield).^[27] Moreover, in comparison to the simple indoline-Tle derived catalyst (**2c**), proline appears to be a key spacer element, likely aiding in transition state organization in analogy to its well-known structure promoting features.^[28]

With the optimized catalysts (**2i** or **2j**) and reaction conditions in hand, we began to explore our main hypothesis in earnest, that these co-catalysts and conditions would provide sufficient handles to adapt to a wide variety of nitrogen-bearing heterocyclic substrates. As shown in Table 3, we were pleased to identify catalysts and conditions that deliver both high enantioselectivity and yield across quite different heterocycle classes, each exhibiting drastic differences in electronic properties, Brønsted basicity and H-bond acceptor capacity.

Notably, as expected, the initially optimized conditions apply to a small set of heterocycles similar to pyridine-3-

Table 3: Scope and limitation.^[a]

Scope:			
 4 , X = H, 99%, 94:6 er [with 2i] 5 , X = OMe, 87%, 96:4 er [with 2i] 6 , X = Me, 84%, 94:6 er [with 2i] 7 , X = Cl, 99%, 95.5:4.5 er [with 2i] 8 , X = CF ₃ , 99%, 94.5:5.5 er [with 2i] 9 , X = CN, 99%, 94.5:5.5 er [with 2i]	 10 , 73%, 95.5:4.5 er [with 2i] 13 ^[b] , 57%, 17:83 er [with 2j]	 11 ^[b] , 97%, 15:85 er [with 2j] 14 ^[b] , 77%, 92.5:7.5 er [with 2j]	 12 ^[b] , 83%, 7:93 er [with 2j] 15 , 88%, 96:4 er [with 2j]
 16 ^[c,d] , 75%, >99.5:0.5 er [with 2j] 17 , 93%, 99:1 er [with 2i] 18 , 89%, 85:15 er [with 2j] 19 ^[c,d] , 85%, 1:99 er [with 2j] 20 ^[c] , 59%, 87:13 er [with 2j]	 21 , 93%, 96:4 er [with 2i] [X-ray structure] 22 , 96%, >99.5:0.5 er [with 2i] [1 gram scale: 97%, >99.5:0.5 er] [X-ray structure] 23 ^[b] , 99%, 97:3 er [with 2j] 24 ^[c] , 81%, 77:23 er [with 2i]	 25 ^[c,d] , 99%, 5:95 er [with 2i] 26 , 99%, 24:76 er [with 2j] 27 , 76%, 12:88 er [with 2j]	 28 ^[c,d,e] , 38%, 52:48 er [with 2i] 29 ^[c] , 82%, 94:6 er [with 2j] [X-ray structure] 30 ^[c,d] , 37%, 74:26 er [with 2j] 31 ^[c,f] , 99%, 88.5:11.5 er [with 2j]
 32 ^[f] , 94%, 92:8 er [with 2j] [X-ray structure] 33 , 87%, 25:75 er [with 2j]	Limitation: 37 , 0% 38 , 0% 39 , 0%		

[a] Reactions were performed at 0.2 mmol of **1** and 0.4 mmol of carboxaldehyde. Isolated yield was shown. Enantiomeric ratio (er) was measured by HPLC employing a chiral stationary phase and is presented in the order of elution in the HPLC chromatogram. The absolute configuration of products **21**, **22**, **29**, **32** was determined by X-ray crystallography and is as shown in the complete structures in the Table. The assignment of absolute stereochemistry of the other compounds by analogy has not been done explicitly due to the diversity of heterocyclic structures presented. [b] 3-CF₃-5-Ar^F-C₆H₃CO₂H (**3b**) was used instead of **3a**. [c] 1,2-dichloroethane as solvent. [d] Reactions were carried at 23 °C. [e] 1 equiv of carboxaldehyde was used. [f] 4-NMe₂-C₆H₄CO₂H (**3c**) was used instead of **3a**. Ar^F = 3,5-CF₃-C₆H₃. Tr = triphenylmethyl.

carboxaldehyde; these very same conditions do not deliver excellent results when the heteroatom is disposed *ortho* to the carbonyl group (see below). Yet, our approach enables quick pivots, and we were gratified to discover that adjusting the pK_a of the carboxylic acid co-catalyst often restored the excellent enantioselectivities for many quite different heteroarenes. That said, some recalcitrant substrate classes were also identified.

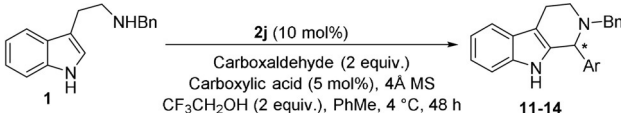
We first examined 6-substituted pyridine-3-carboxaldehydes and found that both electron-donating and withdrawing substrates afforded products in excellent yields and enantioselectivities under the previously optimized conditions (**4–9**; 84–99% yield, 94:6 to 96:4 er). The same was observed for product from pyridine-4-carboxaldehyde (**10**).

A striking difference was observed in the pyridine-2-carboxaldehyde case (**11**). When we used triphenylacetic acid (TrCO_2H , **3a**) as the co-catalyst, the reaction exhibited poor enantioselectivity (41:59 er). However, the more acidic benzoic acid ($3\text{-CF}_3\text{-5-Ar}^F\text{-C}_6\text{H}_3\text{CO}_2\text{H}$, $\text{Ar}^F = 3,5\text{-CF}_3\text{-C}_6\text{H}_3$, **3b**) was found to restore significant enantioselectivity (15:85 er) after extensive screening (See the Supporting Information, Table S3 for details).

This modified system was also applicable to a wide range of heterocycle-substituted aldehydes, including trifluoromethylated pyridine (**12**; 15:85 er), pyrazine (**13**; 17:83 er) and quinoxaline (**14**; 92.5:7.5 er), in which the corresponding carboxaldehyde contained a pyridine-like nitrogen situated *ortho* to the aldehyde (Table 4). It is plausible, although not yet proven, that these compounds bear (*R*)-configuration in analogy to product **47**, whose X-ray crystallographic structure as the methyl ester derivative is shown in Scheme 2 (see below).

Interestingly, TrCO_2H , which was generally found to offer good enantioselectivity as demonstrated not only with pyridines (**4–10**), but also with quinoxaline (**15**; 96:4 er), (iso)quinolines (**16–19**; 85:15 to 99.5:0.5 er) and pyrimidines-substituted cases (**20, 21**; 87:13 er to 96:4 er). A number of these venerable heterocycles were found to exhibit truly

Table 4: Comparison of carboxylic acid cocatalyst for pyridine-2-carboxaldehyde type substrate.^[a]

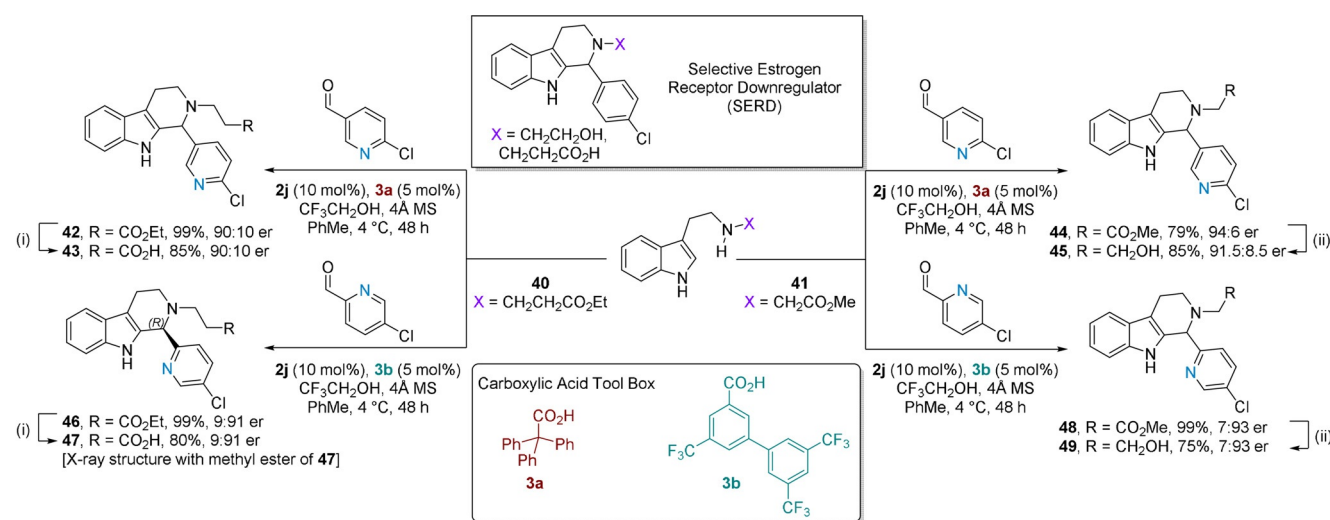


Entry	Carboxylic acid	Corresponding product	er
1	TrCO_2H	11	41:59
2	$3\text{-CF}_3\text{-5-Ar}^F\text{-C}_6\text{H}_3\text{CO}_2\text{H}$	11	15:85
3	TrCO_2H	12	44:56
4	$3\text{-CF}_3\text{-5-Ar}^F\text{-C}_6\text{H}_3\text{CO}_2\text{H}$	12	7:93
5	TrCO_2H	13	42:58
6	$3\text{-CF}_3\text{-5-Ar}^F\text{-C}_6\text{H}_3\text{CO}_2\text{H}$	13	17:83
7	TrCO_2H	14	55:45
8	$3\text{-CF}_3\text{-5-Ar}^F\text{-C}_6\text{H}_3\text{CO}_2\text{H}$	14	92.5:7.5

[a] Reactions were performed at 0.05 mmol scale. Enantioselectivity was measured by HPLC with chiral stationary phase. $\text{Ar}^F = 3,5\text{-CF}_3\text{-C}_6\text{H}_3$.

excellent enantioselectivity (**16, 17, 19**), as illustrated in Table 3. Notably, the absolute configuration of **21** was determined by X-ray crystallography and found to be in the (*R*)-configuration of **21** (i.e., substituent pointing down); that is, opposite that observed for compound **47** (i.e., substituent pointing up),^[29] reflecting as yet unknown mechanistic differences. Accordingly, we also employ extreme caution in the assignment of absolute stereochemistry by analogy throughout this study (see below).

Next, we focused on five-membered ring heterocycles (pyrazole, imidazole, thiazole, isoxazole), with variable substitution patterns, and several of these also exhibit very high enantioselectivity. In the *N*-Ph pyrazole carboxaldehyde series, product **22** from the corresponding 4-carboxaldehyde was obtained in excellent yield (99%) and enantioselectivity



(>99.5:0.5 er), again X-ray crystallography suggested (*R*)-configuration of **22**. The *N*-benzyl group of **22** could be removed by hydrogenolysis and transformed to other functionalities. (See the Supporting Information, Section 4.3 for details). Performing the reaction on gram-scale with reduced catalyst loading (5 mol % **2i**) afforded equally high yield and enantioselectivity. For product from the 3-carboxaldehyde isomer (**23**), a more acidic carboxylic acid **3b** was again identified to be essential for good asymmetric induction (97:3 er; Table 5).

Stimulated by the very high selectivity exhibited by the pyrazole moiety, we examined several substituted isoxazoles (**25–27**). Although isoxazole and pyrazole are isoelectronic five-membered heterocycles, their dependence on the carboxylic acid co-catalyst in PS reaction differed significantly. When the aldehyde was not *ortho* to the heteroatom, a similar trend in enantioselectivity and carboxylic acid co-catalyst dependence were found [**22** (>99.5:0.5 er) vs. **25** (5:95 er); **24** (77:23 er) vs. **27** (12:88 er)].

However, a striking contrast between **23** and **26** was observed in terms of preference in co-catalyst. No improvement in enantiomeric ratio of **26** (24:76 er) was observed by employing a more acidic benzoic acid **3b**. This was contrary to the *N*-Ph pyrazole analogues (**23**) (97:3 er), given the similar topology in **23** and **26**. This discrepancy might come from the electron-withdrawing effect of oxygen on isoxazole, which was reflected from the attenuated basicity of isoxazole (pK_a of conjugate acid = -3 compared to pyrazole pK_a of conjugate acid = $+2.5$).^[8]

Next, imidazole-carboxaldehydes containing *N*-H or *N*-alkyl were compared. When a bulky trityl group (**28**) was present, a nearly racemic product (52:48 er) was obtained. This might suggest that the bulky Tr substituent on the carboxaldehyde disrupts the transition state during the enantiodetermining step. A *N*-Me-bearing substrate (**29**) was obtained in both good yield (82 %) and enantioselectivity

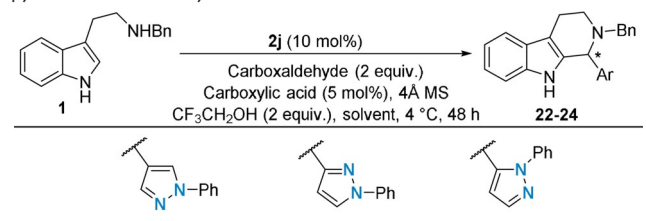
(94:6 er); however, the unprotected *N*-H imidazole (**30**) offered a sluggish reaction (37 %) with only moderate enantioselectivity (74:26 er).

Extending our study to additional medicinally relevant heterocycles, we found that 5- and 2-substituted thiazole carboxaldehydes, with sulfur *ortho* to the aldehyde, gave good results, so long as a less acidic benzoic acid, 4-dimethylaminobenzoic acid (**3c**) ($pK_a=4.8$) was employed for both substrates (**31**, 88.5:11.5 er; **32**, 92:8 er) (Table 6). Notably, product **33** bearing a thiazole 4-carboxaldehyde, was obtained with only moderate enantioselectivity (25:75 er). X-ray crystallographic analysis suggested that the major product **32** has the same configuration presumed for 2-pyridyl type products (i.e., substituent pointing up, by analogy to X-ray crystallographic analysis of methyl ester of **47**, Scheme 2). Lastly, carbolines bearing 7-azaindole (**34**; 3:97 er), pyrrole-pyrimidine (**35**; 5.5:94.5 er) and free indazole (**36**; >99.5:0.5 er) were obtained smoothly under similar conditions. On the other hand, 1,2,4-triazole (**37**), imidazo[1,2-*a*]pyridine (**38**) and piperidine (**39**) were found to be unreactive under the conditions we explored.

As described above, our survey of diverse *N*-heterocyclic carboxaldehydes revealed unusual substrate-selectivity relationships. To gain a deeper insight, we carried out Hammett Analysis for several of the most illustrative aldehydes, namely **4** (Ar = 3-pyridyl), **11** (Ar = 2-pyridyl) and **32** (Ar = 2-thiazolyl), in order to assess a possible correlation between enantioselectivity and acidity of *para*-substituted benzoic acids (Figure 1).^[30]

This experiment revealed a striking level of nuance as the aldehyde is varied. For example, minimal impact on enantioselectivity was observed in the case of **4** as the nature of benzoic acid acidity was altered. In sharp contrast, and in agreement to our initial observation, formation of product **11** (with *ortho* pyridine-type nitrogen) shows a general prefer-

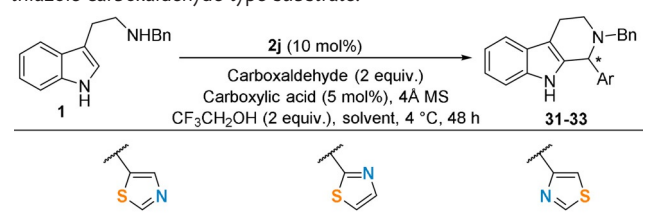
Table 5: Comparison of carboxylic acid cocatalyst for regioisomeric pyrazole carboxaldehyde substrate.^[a]



Entry	Carboxylic acid	Corresponding product	er
1	3-CF ₃ -5-Ar ^F -C ₆ H ₃ CO ₂ H	22	96:4
2	TrCO ₂ H	22	>99.5:0.5
3	4-NMe ₂ -C ₆ H ₄ CO ₂ H	22	98:2
4	3-CF ₃ -5-Ar ^F -C ₆ H ₃ CO ₂ H	23	97:3
5	TrCO ₂ H	23	65:35
6	4-NMe ₂ -C ₆ H ₄ CO ₂ H	23	75:25
7	3-CF ₃ -5-Ar ^F -C ₆ H ₃ CO ₂ H	24	55:45
8	TrCO ₂ H	24	78:22
9	4-NMe ₂ -C ₆ H ₄ CO ₂ H	24	75:25

[a] Reactions were performed at 0.05 mmol scale. Enantioselectivity was measured by HPLC with chiral stationary phase. Ar^F = 3,5-CF₃-C₆H₃. PhMe used as solvent for **22** and **23**. 1,2-DCE used as solvent for **24**.

Table 6: Comparison of carboxylic acid cocatalyst for regioisomeric thiazole carboxaldehyde type substrate.^[a]



Entry	Carboxylic acid	Corresponding product	er
1	3,5-CF ₃ -C ₆ H ₃ CO ₂ H	31	66:34
2	TrCO ₂ H	31	81:19
3	4-NMe ₂ -C ₆ H ₄ CO ₂ H	31	88.5:11.5
4	3,5-CF ₃ -C ₆ H ₃ CO ₂ H	32	47:53
5	TrCO ₂ H	32	74:26
6	4-NMe ₂ -C ₆ H ₄ CO ₂ H	32	92:8
7	3,5-CF ₃ -C ₆ H ₃ CO ₂ H	33	33:64
8	TrCO ₂ H	33	25:75
9	4-NMe ₂ -C ₆ H ₄ CO ₂ H	33	31:69

[a] Reactions were performed at 0.05 mmol scale. Enantioselectivity was measured by HPLC with chiral stationary phase. 1,2-DCE used as solvent for **31**. PhMe used as solvent for **32** and **33**.

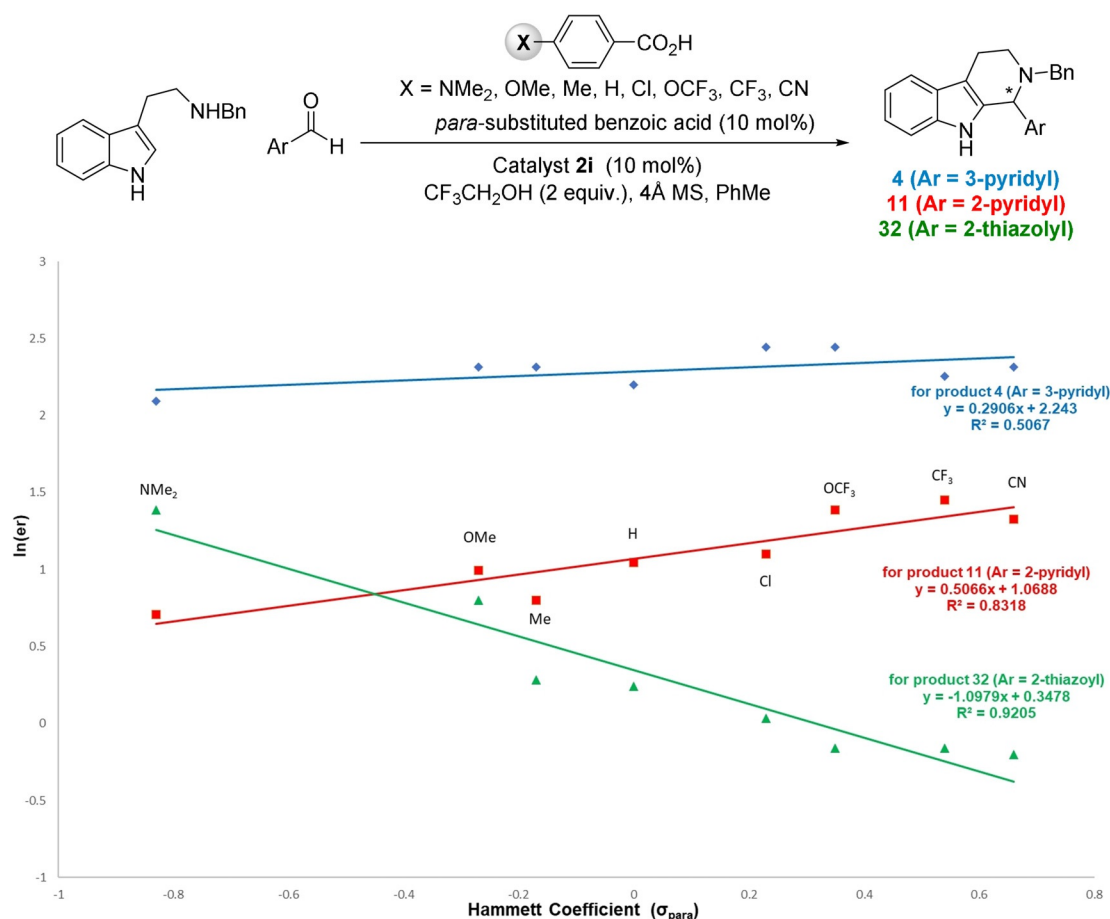


Figure 1. Hammett plot of *para*-substituted benzoic acid and product from different carboxaldehydes.

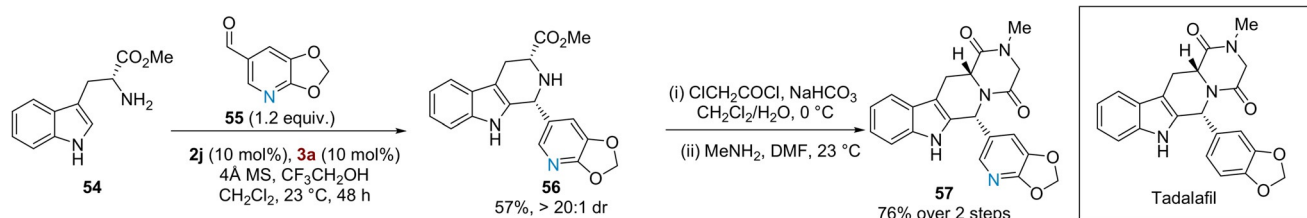
ence for the more acidic benzoic acid; yet, for product **32** (with *ortho* sulfur), the opposite trend is observed, with the best results observed with less acidic benzoic acids. This study highlights and underscores strikingly different behavior as a function of the diverse electronic properties intrinsic to superficially similar heterocycles. Accordingly, we have shown that new and challenging scenarios, wherein substrates that give poor results under standard conditions, can be optimized with this multi-catalytic system, providing better flexibility and tunability for a significantly expanded and broad scope of substrates.

While developing our substrate scope, we noted a series of β -tetrahydrocarbolines prepared and evaluated by AstraZeneca as potent drug candidates (Selective Estrogen Receptor Downregulators, SERD) (Scheme 2) with improved bioavailability for treating advanced estrogen receptor-positive breast cancer.^[22] We were therefore interested to explore the possibility of synthesizing pyridine-containing analogues for these potent molecules as a test of our catalyst system. We found that our methodology worked well with both 6-chloropyridine-3-carboxaldehyde or 5-chloropyridine-2-carboxaldehyde to afford PS products (**42**, **44**, **46**, **48**) in good yield (79–99 %) and enantioselectivity (up to 94:6 er; up to 7:93 er). The careful match between substrate and carboxylic acid co-catalyst was again found to be necessary for achieving high enantioselectivity. The subsequent ester hydrolysis or

reduction afforded SERD pyridine analogues (**43**, **45**, **47**, **49**) with good yield and optical purity. X-ray crystallography suggested the absolute configuration of **47** to be (*R*) (after derivatization into methyl ester).

The tolerance of *N*-substituents other than benzyl (Bn) was thus highlighted in the synthesis of compounds **42**, **44**, **46** and **48**. This finding was further exemplified with *N*-methyl tryptamine **50** (*N*-Me), which was converted to **51** with a 98:2 er and in 67 % yield. Unfortunately, simple tryptamine (**52**) (with free-NH₂) in the place of the *N*-alkyl substituent did not allow for an efficient reaction under the conditions we explored. (See the Supporting Information, Table S9 for details).

Finally, from an experimental perspective, we addressed Tadalafil, a U.S. FDA approved drug for treating erectile dysfunction^[23a] and pulmonary arterial hypertension (Scheme 3).^[23b] We pointed the methodology at the preparation of the pyridine-containing analogue **57**. The key Pictet–Spengler cyclization of D-tryptophan methyl ester (**54**) with carboxaldehyde **55** to product **56** proceeded with excellent diastereoselectivity (> 20:1 dr) in which stereochemistry is dictated by the matching chiral catalyst (See the Supporting Information, Table S6 for details). The desired pyridine-analogue (**57**) of Tadalafil was obtained in good yield and diastereoselectivity after two synthetic steps, and quite notably, without dynamic crystallization.^[23c,d]



Scheme 3. Diastereoselective synthesis of pyridine-containing analogue of Tadalafil. [a] Diastereoselectivity (dr) was measured by ^1H NMR spectroscopy on crude reaction mixture.

Conclusion

We have developed an enantioselective functionalization of diverse aromatic nitrogen heterocycles by means of Pictet–Spengler reaction. Twelve different classes of medically relevant heterocycles were shown to be compatible, although different classes and regioisomers of heterocycles exhibited drastically different reactivity and selectivity, with the position of the heteroatom(s) appearing to play a major role. Enantiodivergence was observed for several substrates that dispose a pyridine-like nitrogen in either the proximal or *ortho*-like position with respect to the aldehyde, versus more distal position. This may be suggestive that two different stereochemical induction models operate under relevant reaction conditions, which requires further study to rationalize in detail. Even so, we have demonstrated that the same chiral catalyst can deliver high enantioselectivities across heterocycle classes, with the use of different achiral carboxylic acid co-catalysts. We believe this demonstrates the benefit of flexibility and tunability offered by a multi-catalytic system when designing reaction systems involving myriad pharmaceutically relevant heterocycles.

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Conflict of Interest

The authors declare no conflict of interest.

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- [1] a) E. Vitaku, D. T. Smith, J. T. Njardarson, *J. Med. Chem.* **2014**, *57*, 10257–10274; b) Novel Drug Approvals for 2020. <https://www.fda.gov/drugs/new-drugs-fda-cders-new-molecular-entities-and-new-therapeutic-biological-products/novel-drug-approvals-2020> (accessed March 5, 2021).
- [2] a) J. A. Joule, K. Mills, *Heterocyclic Chemistry*, 5th ed., Wiley, Hoboken, **2010**; b) T. Eicher, S. Hauptmann, A. Speicher, *The*

Chemistry of Heterocycles, 3rd ed., Wiley-VCH, Weinheim, **2012**; c) I. V. Seregin, V. Gevorgyan, *Chem. Soc. Rev.* **2007**, *36*, 1173–1193; d) M. Baumann, I. R. Baxendale, *Beilstein J. Org. Chem.* **2013**, *9*, 2265–2319; e) A. P. Taylor, R. P. Robinson, Y. M. Fobian, D. C. Blakemore, L. H. Jones, O. Fadeyi, *Org. Biomol. Chem.* **2016**, *14*, 6611–6637.

- [3] a) R. P. Hsung, P.-Q. Huang, Z.-J. Yao, *Efficiency in Natural Product Total Synthesis*, 1st ed., Wiley-VCH, Weinheim, **2018**; b) J. Song, D.-F. Chen, L.-Z. Gong, *Natl. Sci. Rev.* **2017**, *4*, 381–396; c) M. P. van der Helm, B. Klemm, R. Eelkema, *Nat. Rev. Chem.* **2019**, *3*, 491–508; d) M. Odagi, K. Nagasawa, *Asian J. Org. Chem.* **2019**, *8*, 1766–1774; e) B. Han, X.-H. He, Y.-Q. Liu, G. He, C. Peng, J.-L. Li, *Chem. Soc. Rev.* **2021**, *50*, 1522–1586.
- [4] a) M. Taylor, N. Tokunaga, E. N. Jacobsen, *Angew. Chem. Int. Ed.* **2005**, *44*, 6700–6704; *Angew. Chem.* **2005**, *117*, 6858–6862; b) D. A. DiRocco, K. M. Oberg, D. M. Dalton, T. Rovis, *J. Am. Chem. Soc.* **2009**, *131*, 10872–10874; c) J. Douglas, J. E. Taylor, G. Churchill, A. M. Z. Slawin, A. D. Smith, *J. Org. Chem.* **2013**, *78*, 3925–3938; d) D. Best, H. W. Lam, *J. Org. Chem.* **2014**, *79*, 831–845; e) S. Wang, X. Li, H. Liu, J. Zhuang, J. Li, H. Li, W. Wang, *J. Am. Chem. Soc.* **2015**, *137*, 2303–2310; f) X. Bai, G. Zeng, T. Shao, Z. Jiang, *Angew. Chem. Int. Ed.* **2017**, *56*, 3684–3688; *Angew. Chem.* **2017**, *129*, 3738–3742; g) M. Formica, G. Sorin, A. J. M. Farley, J. Díaz, R. S. Paton, D. J. Dixon, *Chem. Sci.* **2018**, *9*, 6969–6974; h) J. S. Möhler, T. Schnitzer, H. Wennemers, *Chem. Eur. J.* **2020**, *26*, 15623–15628.
- [5] a) D.-S. Wang, Q.-A. Chen, S.-M. Lu, Y.-G. Zhou, *Chem. Rev.* **2012**, *112*, 2557–2590; b) A. M. Faísca Phillips, A. J. L. Pombeiro, *Org. Biomol. Chem.* **2017**, *15*, 2307–2340; c) X. Li, J.-J. Tian, X.-S. Tu, N.-N. Zeng, X.-C. Wang, *Angew. Chem. Int. Ed.* **2019**, *58*, 4664–4668; *Angew. Chem.* **2019**, *131*, 4712–4716.
- [6] a) M. Zeng, S.-L. You, *Synlett* **2010**, *9*, 1289–1301; b) R. Dalpozzo, *Chem. Soc. Rev.* **2015**, *44*, 742–778.
- [7] a) S.-Y. Hsieh, Y. Tang, S. Crotti, E. A. Stone, S. J. Miller, *J. Am. Chem. Soc.* **2019**, *141*, 18624–11829; b) E. A. Stone, K. J. Cutrona, S. J. Miller, *J. Am. Chem. Soc.* **2020**, *142*, 12690–12698.
- [8] J. A. Zoltewicz, L. W. Deady, *Adv. Heterocycl. Chem.* **1978**, *22*, 71–121.
- [9] a) C. W. Bird, *Tetrahedron* **1985**, *41*, 1409–1414; b) C. W. Bird, *Tetrahedron* **1986**, *42*, 89–92.
- [10] a) J. Lacour, V. Hebbe-Viton, *Chem. Soc. Rev.* **2003**, *32*, 373–382; b) R. J. Phipps, G. L. Hamilton, F. D. Toste, *Nat. Chem.* **2012**, *4*, 603–614; c) M. Mahlau, B. List, *Angew. Chem. Int. Ed.* **2013**, *52*, 518–533; *Angew. Chem.* **2013**, *125*, 540–556; d) K. Brak, E. N. Jacobsen, *Angew. Chem. Int. Ed.* **2013**, *52*, 534–561; *Angew. Chem.* **2013**, *125*, 558–588.
- [11] For selected examples, see a) S. Lin, E. N. Jacobsen, *Nat. Chem.* **2012**, *4*, 817–824; b) C. S. Yeung, R. E. Ziegler, J. A. Porco, Jr., E. N. Jacobsen, *J. Am. Chem. Soc.* **2014**, *136*, 13614–13617; c) M. Wasa, R. Y. Liu, S. P. Roche, E. N. Jacobsen, *J. Am. Chem. Soc.* **2014**, *136*, 12872–12875; d) S. M. Banik, A. Levina, A. M. Hyde, E. N. Jacobsen, *Science* **2017**, *358*, 761–764; e) A. E. Wendlandt, P. Vangal, E. N. Jacobsen, *Nature* **2018**, *556*, 447–451.

- [12] For selected examples, see: a) S. Das, L. Liu, Y. Zheng, M. W. Alachrag, W. Thiel, C. K. De, B. List, *J. Am. Chem. Soc.* **2016**, *138*, 9429–9432; b) S. Lee, P. S. J. Kaib, B. List, *J. Am. Chem. Soc.* **2017**, *139*, 2156–2159; c) N. Tsuji, J. Kennemur, T. Buyck, S. Lee, S. Prévost, P. S. J. Kaib, D. Bykov, C. Farès, B. List, *Science* **2018**, *359*, 1501–1505; d) T. Gatzemeier, M. Turberg, D. Yepes, Y. Xie, F. Neese, G. Bistoni, B. List, *J. Am. Chem. Soc.* **2018**, *140*, 12671–12676; e) L. Schreyer, P. S. J. Kaib, V. N. Wakchaure, C. Obradors, R. Properzi, S. Lee, B. List, *Science* **2018**, *362*, 216–219.
- [13] For selected examples, see a) Y.-M. Wang, J. We, C. Hoong, V. Rauniyar, F. D. Toste, *J. Am. Chem. Soc.* **2012**, *134*, 12928–12931; b) R. J. Phipps, F. D. Toste, *J. Am. Chem. Soc.* **2013**, *135*, 1268–1271; c) A. J. Neel, J. P. Hehn, P. F. Tripet, F. D. Toste, *J. Am. Chem. Soc.* **2013**, *135*, 14044–14047; d) W. Zi, Y.-M. Wang, F. D. Toste, *J. Am. Chem. Soc.* **2014**, *136*, 12864–12867; e) Z. Yang, Y. He, F. D. Toste, *J. Am. Chem. Soc.* **2016**, *138*, 9775–9778.
- [14] For selected examples, see a) C. Min, N. Mittal, D. X. Sun, D. Seidel, *Angew. Chem. Int. Ed.* **2013**, *52*, 14084–14088; *Angew. Chem.* **2013**, *125*, 14334–14338; b) C. Zhao, S. B. Chen, D. Seidel, *J. Am. Chem. Soc.* **2016**, *138*, 9053–9056; c) C. L. Jarvis, J. S. Hirschi, M. J. Veticatt, D. Seidel, *Angew. Chem. Int. Ed.* **2017**, *56*, 2670–2674; *Angew. Chem.* **2017**, *129*, 2714–2718.
- [15] For selected examples, see a) K. Ohmatsu, M. Kiyokawa, T. Ooi, *J. Am. Chem. Soc.* **2011**, *133*, 1307–1309; b) K. Ohmatsu, Y. Hamajima, T. Ooi, *J. Am. Chem. Soc.* **2012**, *134*, 8794–8797; c) D. Uruguchi, H. Sasaki, Y. Kimura, T. Ito, T. Ooi, *J. Am. Chem. Soc.* **2018**, *140*, 2765–2768.
- [16] For selected examples, see a) D. Nakashima, H. Yamamoto, *J. Am. Chem. Soc.* **2006**, *128*, 9626–9627; b) C. H. Cheong, H. Yamamoto, *J. Am. Chem. Soc.* **2008**, *130*, 9246–9247; c) M. Sickert, C. Schneider, *Angew. Chem. Int. Ed.* **2008**, *47*, 3631–3634; *Angew. Chem.* **2008**, *120*, 3687–3690; d) M. Rueping, B. J. Nachtsheim, S. A. Moreth, M. Bolte, *Angew. Chem. Int. Ed.* **2008**, *47*, 593–596; *Angew. Chem.* **2008**, *120*, 603–606; e) T. Liang, Z. Zhang, J. C. Antilla, *Angew. Chem. Int. Ed.* **2010**, *49*, 9734–9736; *Angew. Chem.* **2010**, *122*, 9928–9930; f) G. Bergonzini, S. Vera, P. Melchiorre, *Angew. Chem. Int. Ed.* **2010**, *49*, 9685–9688; *Angew. Chem.* **2010**, *122*, 9879–9882; g) M. Rueping, U. Uri, M.-Y. Lin, I. Atodiresei, *J. Am. Chem. Soc.* **2011**, *133*, 3732–3735; h) A. G. Schafer, J. M. Wieting, T. J. Fisher, A. E. Mattson, *Angew. Chem. Int. Ed.* **2013**, *52*, 11321–11324; *Angew. Chem.* **2013**, *125*, 11531–11534; i) M. Zurro, S. Asmus, S. Beckendorf, C. Mück-Lichtenfeld, O. G. Mancheño, *J. Am. Chem. Soc.* **2014**, *136*, 13999–14002; j) T. Fischer, J. Bamberger, M. Gómez-Martínez, D. G. Piekarski, O. G. Mancheño, *Angew. Chem. Int. Ed.* **2019**, *58*, 3217–3221; *Angew. Chem.* **2019**, *131*, 3250–3255.
- [17] a) T. V. Sravanthi, S. L. Manju, *Eur. J. Pharm. Sci.* **2016**, *91*, 1–10; b) N. Chadha, O. Silakari, *Eur. J. Med. Chem.* **2017**, *134*, 159–184; c) S. He, Z. Lai, Z. Ye, P. H. Dobbelaar, S. K. Shah, Q. Truong, W. Du, L. Guo, J. Liu et al., *ACS Med. Chem. Lett.* **2014**, *5*, 748–753; d) V. Gorki, N. S. Walter, R. Singh, M. Chauhan, N. Dhingra, D. B. Salunke, S. Kaur, *ACS Omega* **2020**, *5*, 17993–18006; e) L. P. Cameron, R. J. Tombari, J. Lu, A. J. Pell, Z. Q. Hurley, Y. Ehinger, M. V. Vargas, M. N. McCarroll, J. C. Taylor, D. Myers-Turnbull et al., *Nature* **2021**, *589*, 474–479.
- [18] a) F. Lovering, J. Bikker, C. Humblet, *J. Med. Chem.* **2009**, *52*, 6752–6756; b) S. Caille, S. Cui, M. M. Faul, S. M. Mennen, J. S. Tedrow, S. D. Walker, *J. Org. Chem.* **2019**, *84*, 4583–4603.
- [19] During the submission process of this manuscript, two synthetic methodologies with the focus on catalytic asymmetric functionalization of diverse *N*-heterocyclic substrates have been reported: a) X. Wang, J. Ke, Y. Zhu, A. Deb, Y. Xu, X. P. Zhang, *J. Am. Chem. Soc.* **2021**, *143*, 11121–11129; b) J. C. Sharland, B. Wei, D. J. Hardee, T. R. Hodges, W. Gong, E. A. Voight, H. M. L. Davies, *Chem. Sci.* **2021**, *12*, 11181–11190.
- [20] For selected examples and reviews in organocatalytic and biocatalytic Pictet–Spengler Reaction, see: a) M. S. Taylor, E. N. Jacobsen, *J. Am. Chem. Soc.* **2004**, *126*, 10558–10559; b) J. Seayad, A. M. Seayad, B. List, *J. Am. Chem. Soc.* **2006**, *128*, 1086–1087; c) M. J. Wanner, R. N. S. van der Haas, K. R. de Cuba, J. H. van Maarseveen, H. Hiemstra, *Angew. Chem. Int. Ed.* **2007**, *46*, 7485–7487; *Angew. Chem.* **2007**, *119*, 7629–7631; d) N. V. Sewgobind, M. J. Wanner, S. Ingemann, R. de Gelder, J. H. van Maarseveen, H. Hiemstra, *J. Org. Chem.* **2008**, *73*, 6405–6408; e) M. E. Muratore, C. A. Holloway, A. W. Pilling, R. I. Storer, G. Trevitt, D. J. Dixon, *J. Am. Chem. Soc.* **2009**, *131*, 10796–10797; f) R. S. Klausen, E. N. Jacobsen, *Org. Lett.* **2009**, *11*, 887–890; g) J. Stöckigt, A. P. Antonchick, F. Wu, H. Waldmann, *Angew. Chem. Int. Ed.* **2011**, *50*, 8538–8563; *Angew. Chem.* **2011**, *123*, 8692–8719; h) D. Huang, F. Xu, X. Lin, Y. Wang, *Chem. Eur. J.* **2012**, *18*, 3148–3152; i) N. Mittal, D. X. Sun, D. Seidel, *Org. Lett.* **2014**, *16*, 1012–1015; j) S.-G. Wang, Z.-L. Xia, R.-Q. Xu, X.-J. Liu, C. Zheng, S.-L. You, *Angew. Chem. Int. Ed.* **2017**, *56*, 7440–7443; *Angew. Chem.* **2017**, *129*, 7548–7551; k) L. Qi, H. Hou, F. Ling, W. Zhong, *Org. Biomol. Chem.* **2018**, *16*, 566; l) R. Andres, Q. Wang, J. Zhu, *J. Am. Chem. Soc.* **2020**, *142*, 14276–14285; m) J. F. Treimer, M. H. Zenk, *Eur. J. Biochem.* **1979**, *101*, 225–233; n) P. Bernhardt, A. R. Usera, S. E. O'Connor, *Tetrahedron Lett.* **2010**, *51*, 4400–4402; o) D. Pressnitz, E.-M. Fischereder, J. Pletz, C. Kofler, L. Hammerer, K. Hiebler, H. Lechner, N. Richter, E. Eger, W. Kroutil, *Angew. Chem. Int. Ed.* **2018**, *57*, 10683–10687; *Angew. Chem.* **2018**, *130*, 10843–10847; p) A. Calcaterra, L. Mangiardi, G. D. Monache, D. Quaglio, S. Balducci, S. Berardozi, A. Iazzetti, R. Franzini, B. Botta, F. Ghirga, *Molecules* **2020**, *25*, 414–496.
- [21] For selected reviews of peptide as catalyst, see: a) E. A. C. Davie, S. M. Mennen, Y. Xu, S. J. Miller, *Chem. Rev.* **2007**, *107*, 5759–5812; b) “Peptides as Asymmetric Organocatalysts”: A. Fingerhut, D. Grau, S. B. Tsogoeva in *Sustainable Catalysis* (Eds.: M. North), RCS, London, **2015**, pp. 309–353; c) K. Akagawa, K. Kudo, *Acc. Chem. Res.* **2017**, *50*, 2429–2439; d) A. J. Metrano, C. R. Shugrue, B. Kim, A. J. Chinn, E. A. Stone, S. J. Miller, *Chem. Rev.* **2020**, *120*, 11479–11615.
- [22] C. De Savi, R. H. Bradbury, A. A. Rabow, R. A. Norman, C. de Almeida, D. M. Andrews, P. Ballard, D. Buttar, R. J. Callis, G. S. Currie, J. O. Curwen, C. D. Davies, C. S. Donald, L. J. L. Feron, H. Gingell, S. C. Glossop, B. R. Hayter, S. Hussain, G. Karoutchi, S. G. Lamont, P. MacFaul, T. A. Moss, S. E. Pearson, M. Tonge, G. E. Walker, H. M. Weir, Z. Wilson, *J. Med. Chem.* **2015**, *58*, 8128–8140.
- [23] a) R. M. Coward, C. C. Carson, *Ther. Clin. Risk Manage.* **2008**, *4*, 1315–1329; b) C. C. Carson, M. Rosenberg, J. Kissel, D. G. Wong, *Int. J. Clin. Pract.* **2014**, *68*, 94–103; c) P. J. Dunn, *Org. Process Res. Dev.* **2005**, *9*, 88–97; d) X.-X. Shi, S.-L. Liu, W. Xu, Y.-L. Xu, *Tetrahedron: Asymmetry* **2008**, *19*, 435–442.
- [24] For reviews in additive, including alcohol in asymmetric catalysis, see: a) T. Sugiishi, M. Matsugi, H. Hamamoto, H. Amii, *RSC Adv.* **2015**, *5*, 17269–17282; b) L. Hong, W. Sun, D. Yang, G. Li, R. Wang, *Chem. Rev.* **2016**, *116*, 4006–4123.
- [25] For selected examples of non-covalent interaction (π - π) between substrates and chiral hydrogen-bond donor catalysts, see: a) R. R. Knowles, S. Lin, E. N. Jacobsen, *J. Am. Chem. Soc.* **2010**, *132*, 5030–5032; b) C. Uyeda, E. N. Jacobsen, *J. Am. Chem. Soc.* **2011**, *133*, 5062–5075; c) A. R. Brown, C. Uyeda, C. A. Brotherton, E. N. Jacobsen, *J. Am. Chem. Soc.* **2013**, *135*, 6747–6749; d) G. Su, C. J. Thomson, K. Yamazaki, D. Rozsar, K. E. Christensen, T. A. Hamlin, D. J. Dixon, *Chem. Sci.* **2021**, *12*, 6064–6072; e) E. Matador, J. Iglesias-Sigüenza, D. Monge, P. Merino, R. Fernández, J. M. Lassaletta, *Angew. Chem. Int. Ed.* **2021**, *60*, 5096–5101; *Angew. Chem.* **2021**, *133*, 5156–5161.

- [26] For a recent review of interaction between substrate and amide's oxygen in peptide, see: A. J. Metrano, S. J. Miller, *Acc. Chem. Res.* **2019**, *52*, 199–215.
- [27] For a literature precedent with indoline as C-termini in amino acid derived bis-thiourea, see: Y. Park, K. C. Harper, N. Kuhl, E. Kwan, R. Y. Liu, E. N. Jacobsen, *Science* **2017**, *355*, 162–166.
- [28] For examples of relationship between peptidic catalyst activity and conformation, see: a) A. J. Metrano, N. C. Abascal, B. Q. Mercado, E. K. Paulson, A. E. Hurtley, S. J. Miller, *J. Am. Chem. Soc.* **2017**, *139*, 492–516; b) X. C. Yan, A. J. Metrano, M. J. Robertson, N. C. Abascal, J. Tirado-Rives, S. J. Miller, W. L. Jorgensen, *ACS Catal.* **2018**, *8*, 9968–9979; c) C. Rigling, J. K. Kisunzu, J. Duschmalé, D. Häussinger, M. Wiesner, M.-C. Ebert, H. Wennemers, *J. Am. Chem. Soc.* **2018**, *140*, 10829–10838.
- [29] For selected examples in substrate controlled enantiodivergence, see: a) J. Wang, J. Chen, C. W. Kee, C.-H. Tan, *Angew. Chem. Int. Ed.* **2012**, *51*, 2382–2386; *Angew. Chem.* **2012**, *124*, 2432–2436; b) K. Mori, T. Itakura, T. Akiyama, *Angew. Chem. Int. Ed.* **2016**, *55*, 11642–11646; *Angew. Chem.* **2016**, *128*, 11814–11818; c) A. L. Featherston, C. R. Shugrue, B. Q. Mercado, S. J. Miller, *ACS Catal.* **2019**, *9*, 242–252.
- [30] For examples of Hammett plot of relationship between electronic property of (co)-catalyst and enantioselectivity, see: a) M. Palucki, N. S. Finney, P. J. Pospisil, M. L. Güler, T. Ishia, E. N. Jacobsen, *J. Am. Chem. Soc.* **1998**, *120*, 948–954; b) K. H. Jensen, M. S. Sigman, *J. Org. Chem.* **2010**, *75*, 7194–7201; c) C. Yang, E.-G. Zhang, X. Li, J.-P. Cheng, *Angew. Chem. Int. Ed.* **2016**, *55*, 6506–6510; *Angew. Chem.* **2016**, *128*, 6616–6620; d) Y.-C. Chan, X. Wang, Y.-P. Lam, J. Wong, Y.-L. S. Tse, Y.-Y. Yeung, *J. Am. Chem. Soc.* **2021**, *143*, 12745–12754.

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